IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – Smart Study Generator Agent

Domain of Project –Education Technology (EdTech)

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Problem Statement -

Problem Statement: Smart Study Generator Agent
The Challenge
Students often struggle with scattered learning resources such as lecture notes, textbooks, research papers, and online tutorials. These resources are difficult to organize, summarize, and personalize, making exam preparation time-consuming and inefficient.

The Objective
Aggregate learning resources into structured study material.
Generate summaries, flashcards, and concept maps.
Create personalized study plans.
Recommend important exam topics.
Track learning progress and performance.

Proposed Solution -

Proposed Solution: AI-Powered Smart Study Generator Agent
The proposed solution is an **Agentic AI-powered Smart Study Generator** developed using **IBM watsonx Orchestrate**, **IBM Granite Models**, and **Retrieval-Augmented Generation (RAG)**.
The system collects learning content from lecture notes, textbooks, PDFs, and online resources through a knowledge base. Using RAG, it retrieves relevant information and generates concise summaries, flashcards, concept maps, quizzes, and personalized study plans.

The solution also analyzes student learning patterns, identifies weak areas, recommends important topics, and provides intelligent study assistance through an interactive conversational interface.
Overall, the Smart Study Generator acts as an intelligent learning companion that simplifies studying, improves productivity, and enhances exam preparation.

Technology Used

- **Langflow platform** -For designing and orchestrating multi-agent workflows visually

- **IBM BOB (Business Operations Builder)** – Used to build, configure, and orchestrate the Smart Study Generator Agent through a no-code/low-code platform.

- **IBM Granite Foundation Models** – For natural language understanding, summarization, question answering, and personalized study assistance.

- **IBM Cloud** – Provides the infrastructure to build, deploy, and securely manage the AI-powered study agent.

- **Retrieval-Augmented Generation (RAG)** – Retrieves relevant study materials from the knowledge base before generating accurate and context-aware responses
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- **Knowledge Base** – Stores lecture notes, textbooks, PDFs, and other learning resources for intelligent retrieval.

- **Prompt Engineering** – Defines the agent's behavior, response format, and study guidance rules.
- **Agentic AI** – Enables autonomous reasoning, personalized learning, and intelligent decision-making for adaptive study support.

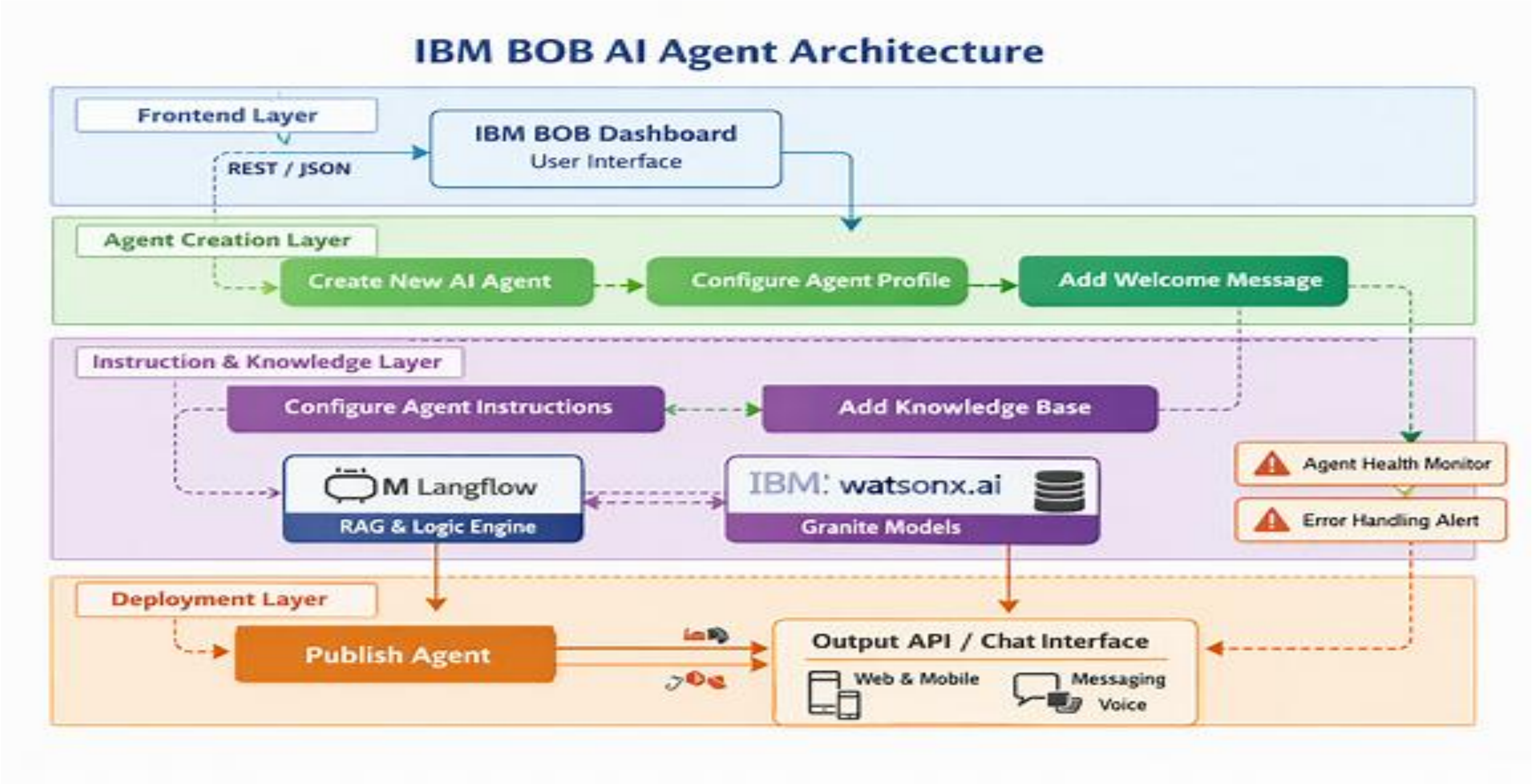
IBM BOB Components Used -

1. **Chat Interface** – Allows students to interact with the Smart Study Generator.
2. **IBM BOB Agent** – Processes user requests and orchestrates the AI workflow.
3. **IBM Granite Foundation Model** – Generates summaries, quizzes, flashcards, and personalized study plans.
4. **Knowledge Base** – Stores lecture notes, textbooks, PDFs, and study materials.
5. **Behavior Instructions** – Controls the agent's responses and study workflow.
6. **Quick Start Prompts** – Provides predefined study-related prompts for faster interaction.

Workflow -


Start
IBM BOB
AI Agent Build
Workflow
Login to
IBM BOB
IBM BOB
tAI Agent
build Morkflow.
Create A
New AI Agent
Start a new
agent project
Configure
Agent Profile
Define
metadata
Add
Welcome
Message
Set a greeting
message
Configure
Agent
Instructions
Add system
prompts
Publish
the Agent
Deploy the
agent
Done

Architecture blueprint



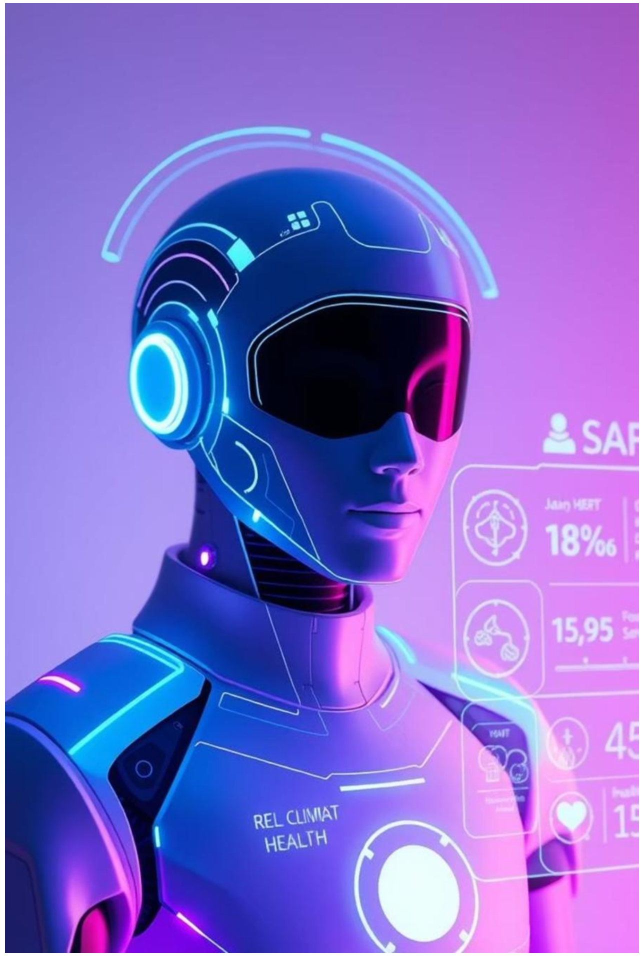
Role of Agentic AI in the solution

Role of Agentic AI

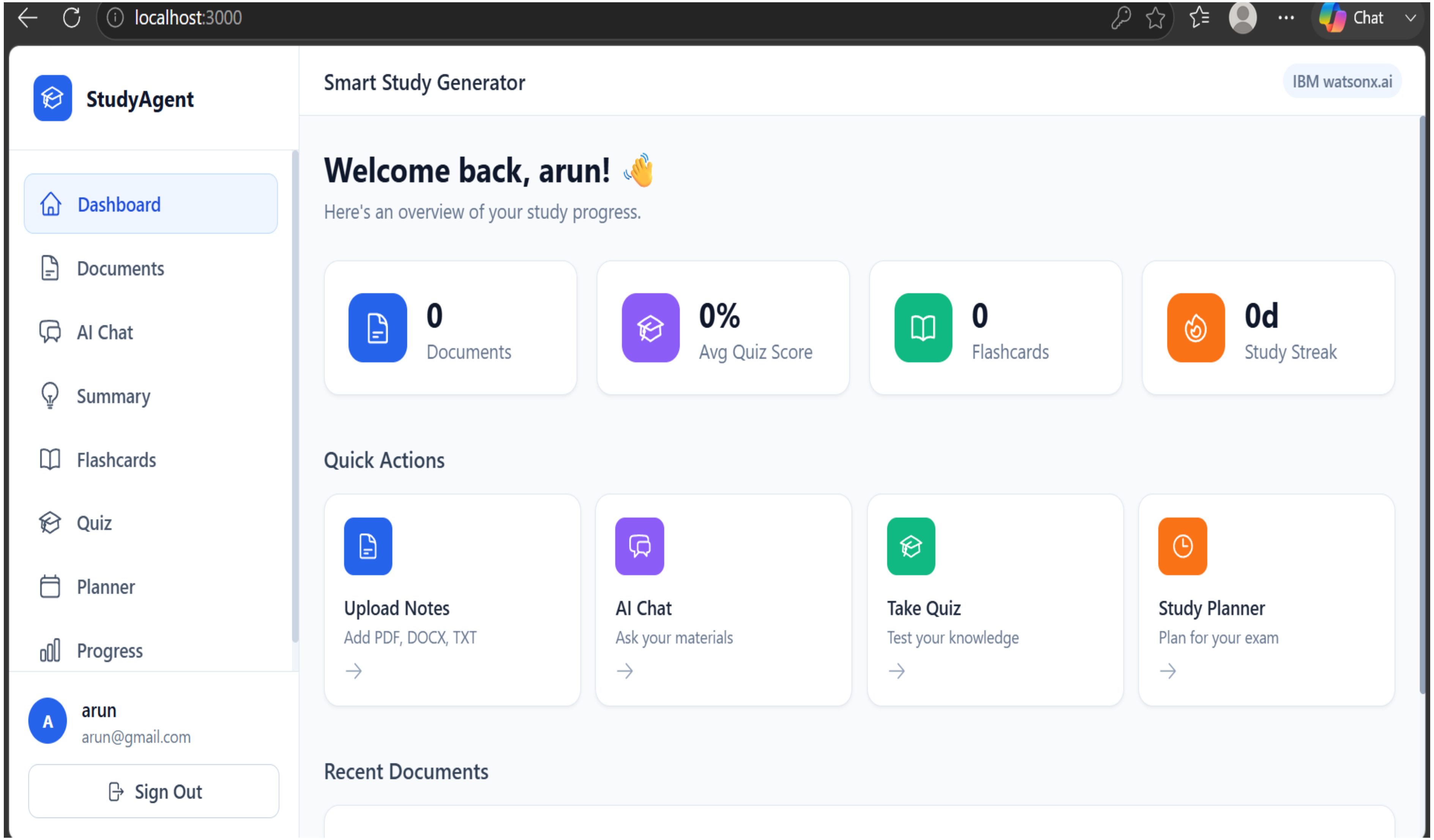
Agentic AI enables the Smart Study Generator to function as an intelligent learning assistant rather than a simple chatbot. Built using **IBM BOB**, **IBM Granite Models**, and **RAG**, the agent retrieves educational content from the knowledge base and generates personalized summaries, flashcards, quizzes, and study plans.

The system understands learning needs, identifies weak areas, recommends important topics, and provides context-aware study guidance. This makes learning more personalized, efficient, and interactive.

Overall, Agentic AI transforms the system into a proactive digital tutor that enhances student learning and exam preparation..



Output


localhost:3000
Chat
StudyAgent
Smart Study Generator
IBM watsonx.ai
Dashboard
Documents
AI Chat
Summary
Flashcards
Quiz
Planner
Progress
arun
arun@gmail.com
Sign Out
Welcome back, arun!
Here's an overview of your study progress.
0
Documents
0%
Avg Quiz Score
0
Flashcards
0d
Study Streak
Quick Actions
Upload Notes
Add PDF, DOCX, TXT
AI Chat
Ask your materials
Take Quiz
Test your knowledge
Study Planner
Plan for your exam
Recent Documents

Novelty and Uniqueness

The proposed **Smart Study Generator Agent** stands out by combining **Agentic AI**, **RAG**, and **personalized learning** into a single intelligent study assistant.

Intelligent Study Assistance – Instead of a static study tool, the agent provides dynamic, context-aware learning support tailored to each student's needs.

RAG-Based Accuracy – Retrieves relevant information from uploaded study materials before generating responses, ensuring accurate and reliable study content.

Personalized Learning – Creates customized study plans, summaries, flashcards, and quizzes based on the student's learning goals and progress.

Multimodal Learning Support – Accepts text, PDFs, images, and voice queries to make studying more interactive and flexible.

Predictive Study Guidance – Recommends important topics, highlights weak areas, and suggests exam-focused preparation strategies.

Real-Time Progress Tracking – Monitors learning progress, practice performance, and study milestones to improve exam readiness.

Scalable & Enterprise-Ready – Built using **IBM BOB**, **IBM Granite Foundation Models**, and **IBM Cloud** for reliable, secure, and scalable deployment.

This makes the solution a smart, adaptive, and proactive digital study companion, not just a traditional note-taking or summarization tool.

Git Hub Link

Please create public github repository (Enable Add README button while creating repository)

with format –

Github URL –https://github.com/krishnanjalitm/smart-study-agent

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Future Scope

1. AI-Powered Voice Assistant & Multilingual Learning –

The system can be enhanced with a voice-enabled assistant and multilingual support, allowing students to interact naturally in their preferred language. This will improve accessibility, hands-free learning, and make the platform more inclusive for diverse learners.

2. Adaptive Learning Analytics & Predictive Performance –

The system can analyze students' learning patterns, quiz performance, and study progress to predict exam readiness, identify weak areas, and recommend personalized study plans. This will enable smarter learning strategies and improve academic outcomes through data-driven insights.

Certificates Earned:

Credly certificate

In recognition of the commitment to achieve professional excellence

Getting Started with Artificial Intelligence
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IBM

Krishnanjali T M

Getting Started with Artificial Intelligence
Has successfully satisfied the requirements for:

Issued on:Jul06, 2026
Issued by:IBM SkillsBuild

Verify: https://www.credly.com/badges/553dd6f2-db47-4aae-9942-619fab3eef82

IBM

Certificates Earned:

IBM BOB Certificate

IBM **SkillsBuild** Completion Certificate

This certificate is presented to

Krishnanjali T M

for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 07 Jul 2026 (GMT)

Learning hours: 30 mins

Thank You!

Thank you for your time and interest.